

Functional Programming

Lab 03: Data Processing Using Python

V. Markos

Week 17, Spring, 2025

Abstract

This lab aims to explore some key data mining concepts using Python and its functional perks. We are going to start small with a snapshot of a books' dataset and then proceed to have actual fun with a real goodreads dataset!

1 Warm Up

Consider the data found into `./source/books.json`, which contains a few well-known books and relevant data about them. We will use this small snapshot as a basis upon which we are going to test some functional approaches on data processing. In what follows, you are strongly encouraged to avoid using loops, while you are required to ensure that your functions have no side-effects¹.

Exercise 1.1 (Filtering). Implement the following functions:

1. Write a function using `filter` and an appropriate lambda function to find all books published before 1960. Then, generalise your implementation to find all books before a date provided as an argument.
2. Write a function using `filter` and a lambda function to find all books in the “Science Fiction” genre. Then, generalise your implementation to find all books within a certain genre, provided as an argument.
3. Write a function using `filter` to return all books with a rating of 4.5 or higher. Then, generalise your implementation to find all books above a certain rating, provided as an argument.

Exercise 1.2 (Mapping). Implement the following functions:

1. Write a function using `map` and a lambda function to create a list of book titles.
2. Write a function using `map` and a lambda function to create a list of author names.
3. Write a function using `map` to create a new list of dictionaries that only contain the book titles and authors.

¹Evidently, I/O and exceptions break this pattern in Python, but, besides those cases, your functions should not generate any other side effects.

Okay, now that we have gotten a taste of functional Python bits, let's try some more complex stuff:

Exercise 1.3. Combining `map` and `filter`:

1. Find the titles of all “Fantasy” books published after 1950.
2. Find the names of all authors that wrote books with a rating above 4.6.

Another common functional pearl is the **reduction** operation, quite similar to folding, as we have seen during our time with Haskell. So...

Exercise 1.4 (Reducing). Using `functools.reduce`:

1. Calculate the average of all books in the dataset.
2. Find the book title with the oldest publication year.

You are not allowed to use loops in none of the above tasks. Think of clever ways to utilise `functools.reduce` to achieve the required functionality in each case.

One last functional programming pearl we have not utilised so far are higher level functions:

Exercise 1.5 (Higher Level Functions). Define a function `apply_criteria` that takes as arguments a list of books as the one provided above and a criteria function, i.e., a boolean function that decides whether a book is desirable or not, and returns the list of all entries matching those criteria.

2 Actual Fun

In the previous section we worked with a dummy dataset, focusing mostly onto getting accustomed with some functional aspects of Python. We are now ready to play with a more realistic dataset, which you can find here:

`./source/archive.zip`.

Exercise 2.1 (Exploratory Data Analysis). Write appropriate Python functions to compute the following quantities and statistical functions:

- the absolute and relative frequency of occurrence of a certain / each value within a list of (potentially categorical) values;
- the average value of a list of numerical values;
- the extreme (max and min) values of a list of numerical values;
- the median of a list of numerical values;
- the standard deviation of a list of numerical values.

Then use those functions to define a function, `eda`, that prints all relevant information for each column of the provided dataset. Feel free to improvise on the format of the output, but stay functional and avoid impurities, side-effects² and anything else Haskell avoids by its nature!

Having a brief description of our dataset, it is high time we drew some conclusions based on that. Evidently, we cannot be fully confident about our inferences just based on a single source of data, however we can get some pointers for future research:

Exercise 2.2. Consider the following questions:

- Q1. Does a book's publisher determine its fate? In other words, is it really important to get published under a good publication house or a book's merits suffice?
- Q2. Are bigger books better? Does page count affect the quality of a book, as perceived by its audience?

Address them both developing any Python functions you see fit for the purpose of each question. You might, among others, use plots, tables, and any other tools you might find helpful in gaining a better understanding for each question separately.

²You might, however, allow yourselves some typical imperative style code only for loading the file and maybe pre-processing it a little bit, if you find this fit for your purpose.